

MOHIT MISRA

Full Stack GenAI / Agentic AI Developer · Python / FastAPI · React / Next.js · RAG, Agents & LLM Systems (LangChain · LangGraph · MCP)

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CORE SKILLS

GenAI / LLM: RAG & retrieval pipelines (hybrid keyword + vector), RAG strategies, LLM orchestration, AI agents, tool / function calling, LangChain, LangGraph, MCP (Model Context Protocol), structured outputs (JSON), Anthropic Claude API, prompt engineering & prompt design, LLM vision extraction, semantic search, cosine-similarity ranking, chunking

Backend: Python (async), FastAPI, Pydantic, Node.js, Express.js, REST API design, JWT auth, role-based access control (RBAC), SSE streaming, scalable system design & architecture

Data: PostgreSQL + pgvector (HNSW), MongoDB, vector databases, embedding pipelines

Cloud & DevOps: AWS (S3 object storage, EC2, RDS, SES, Route 53), CI/CD with GitHub Actions, Docker & Docker Compose, cloud deployment

Frontend: React.js, Next.js, Vite, Tailwind CSS, responsive UI

Tools: Git, GitHub, Postman, Meta Pixel, VS Code

IoT: ESP32 / ESP8266, MQTT, HTTP, sensor data pipelines, Alexa Skills, Google Assistant

EDUCATION

B.Tech, Computer Science & Engineering
National Institute of Science & Technology (NIST), BPUT
2017 – 2021 · CGPA 9.06

CERTIFICATIONS

RAG and Agentic AI Professional Certificate — IBM (verified)
RAG systems, AI agents, LangChain, LangGraph, vector databases, multi-agent workflows

PUBLICATIONS

› Two-Wheeler Anti-Theft System Based on Three-Layer Architecture — Springer
Singapore, 2022

› RF-Based Social Distance Smart Band System — Apple Academic Press, 2022

ACHIEVEMENTS

› Finalist (Top 10) — CSI In-App Student Project Awards 2020

› Top-9 Teams — NASA Space Apps Challenge 2019

› Certificate of Excellence, NIST — SGPA 9.21

SUMMARY

Full Stack GenAI / Agentic AI developer with 5+ years of software engineering and startup experience building production AI systems: a streaming RAG + tool-calling chatbot (FastAPI · Anthropic Claude · pgvector), an LLM-vision structured-extraction pipeline, and embeddings-based semantic search. Startup co-founder & CTO; led an 8-engineer team from data ingestion and vector retrieval to React frontends, AWS deployment, and CI/CD.

PROFESSIONAL EXPERIENCE

Co-Founder, CTO & Director · AROSCAI Technologies Pvt. Ltd.

Nov 2021 – Present | aroscai.com

- Architected a production GenAI support chatbot (FastAPI + Anthropic Claude): SSE streaming, hybrid RAG (keyword/tag scoring + pgvector cosine-similarity search, HNSW index) and bounded tool-calling for live warranty lookups.
- Built the knowledge-ingestion pipeline — PDF/HTML extraction, paragraph-aware chunking, local embeddings (no GPU) — with an admin tool letting non-engineering staff update the knowledge base without code changes.
- Shipped AI Smart Estimate: switchboard photo → structured LLM-vision extraction → automatic catalog matching → lead-gen cart, with a deterministic no-LLM re-match path that avoids repeat inference costs on user edits.
- Developed a role-based ERP/CRM (React + Vite, JWT, per-module permissions): leads, tickets, installations, B2B hierarchies, catalog, and chatbot-conversation review.
- Deployed and operated infrastructure on AWS (EC2, RDS, S3, SES, Route 53) with Docker-based builds and a proposed CI/CD pipeline on GitHub Actions.
- Implemented growth infrastructure: 19 programmatic SEO landing pages, JSON-LD, dynamic sitemap/OG images, and Meta Pixel conversion tracking.
- Led an 8-engineer team on the IoT smart-home platform; shipped Alexa & Google Assistant voice integrations in-house and directed an outsourced partner through the iOS/Android companion apps.

Full Stack AI Developer · VLSIPRO Technologies Pvt. Ltd.

Mar 2025 – Jul 2026 | Contract engagement (completed)

- Built TalentTrack, a MERN recruitment platform with semantic candidate search: resume-embedding pipeline + cosine-similarity ranking matching candidates to jobs by meaning, not keywords.
- Designed REST APIs for recruiter dashboards, job postings, and candidate indexing/retrieval workflows.
- Delivered Knowledge Spark LMS (knowledgespark.in) on WordPress + Tutor LMS: course management, automated enrollment, payment gateway.

Digital Design Engineer · Marquee Semiconductor Pvt. Ltd.

Dec 2020 – Nov 2021

- Built IoT data-acquisition systems (ESP32/ESP8266) and backend logging pipelines streaming real-time sensor data to cloud servers; designed API integrations between embedded hardware and web apps.
- Set the foundations for an internal project-management tool that later grew in scale into corplink.ai, an independent startup entity.

PERSONAL PROJECTS

AgentDesk — Multi-Agent Support Assistant

- Orchestrating specialized agents (retrieval, ticketing, escalation) with LangGraph state machines and LangChain tool-calling; internal tools (order lookup, ticket creation) exposed via MCP servers for dynamic agent discovery.
- Grounding responses with pgvector semantic retrieval and reranking over a hybrid keyword + vector knowledge corpus.

DocMind — MCP-Native Document Research Agent

- Multi-step reasoning agent using LangGraph state graphs and LangChain retrieval chains for iterative document research.
- MCP server exposing search, summarization, and citation tools consumable by any MCP-compatible client; hybrid retrieval (vector + keyword) with structured JSON outputs.